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Enterprise Wireless Network Security: WPA2, MAC Filtering & Router Hardening (Linksys EA6200)

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Lab Overview

This lab demonstrates wireless router security hardening techniques using the Linksys EA6200 Smart Wi-Fi router simulator. The original lab manual specifies the WRT54G legacy router, but the EA6200 was used instead as it provides a more modern and comprehensive interface while covering all the same security objectives.

Lab Objectives

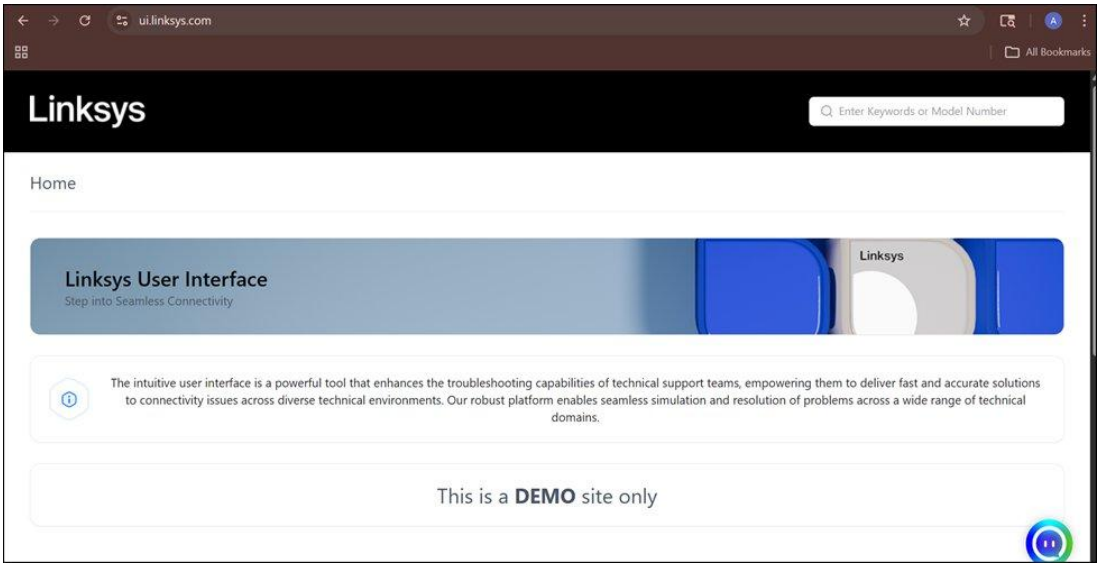
- Configure WPA2 Personal encryption on both 2.4 GHz and 5.0 GHz bands
- Enable MAC Address filtering in whitelist mode
- Disable WPS (Wi-Fi Protected Setup) to remove vulnerabilities
- Disable the Guest Network to prevent unauthorized access
- Change the Router Admin Password to a strong credential
- Verify Firewall settings (IPv4/IPv6 SPI, Internet filters)
- Disable DMZ and DDNS for a minimal attack surface
- Enable Router Logs for security audit trail

Router Used

- **Linksys EA6200** — Smart Wi-Fi Dual-Band Router
- **http://ui.linksys.com** (Linksys User Interface Demo Site)
- **1.1.40.153121**

Step 1: Navigate to the Router Simulator

Open Google Chrome and go to http://ui.linksys.com — this is the official Linksys UI demo site used for lab practice.

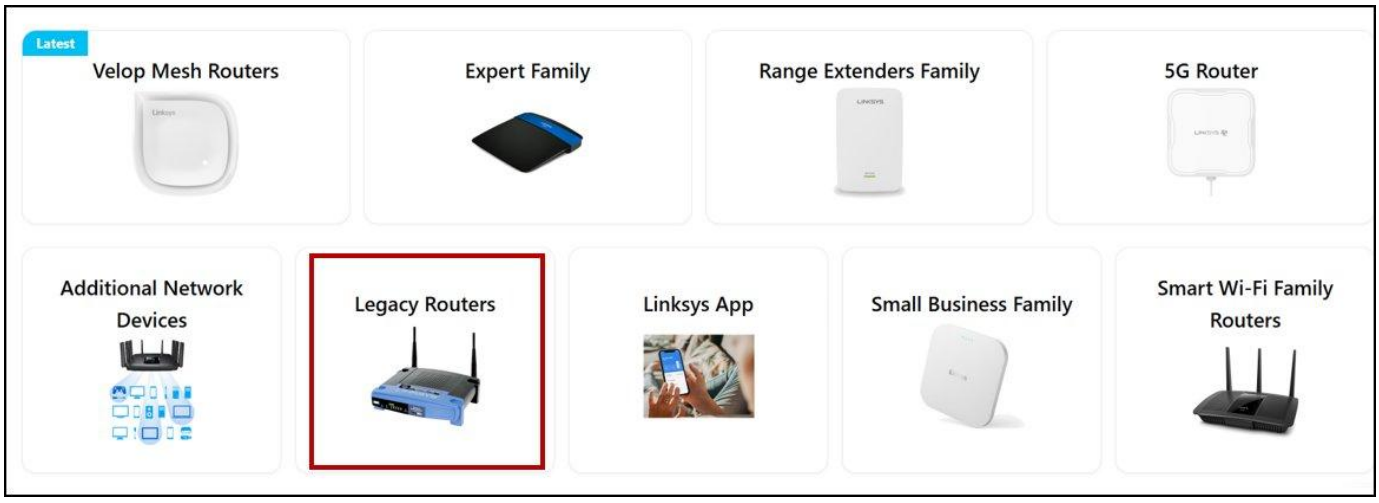


Linksys User Interface demo home page at ui.linksys.com

- **Open Chrome browser → type http://ui.linksys.com → press Enter**
- **You will see "This is a DEMO site only" — this is correct and expected**

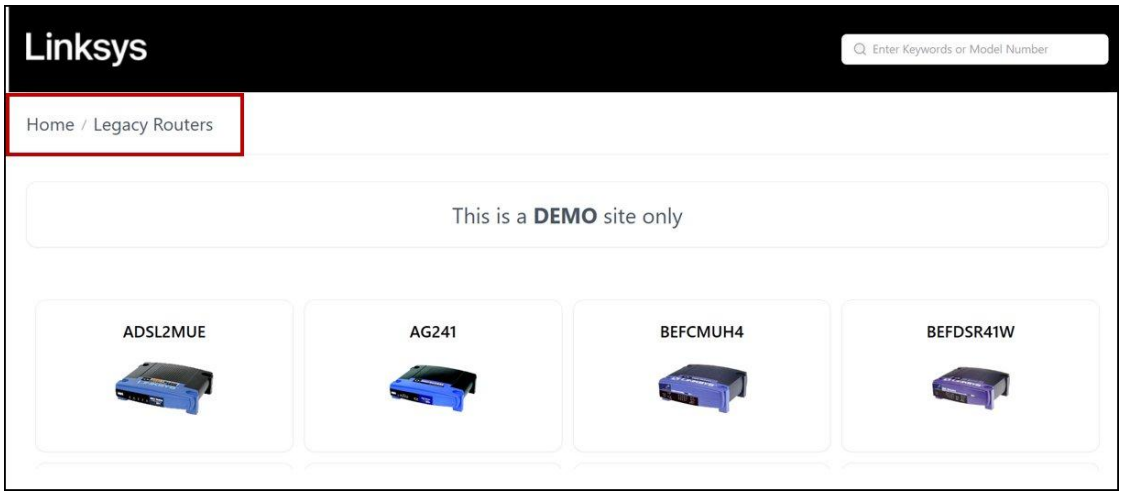
Navigate to Legacy Routers

On the home page, scroll down to find the router family categories. Click on "Legacy Routers" to access the EA6200 simulator.



Click on "Legacy Routers" section (highlighted in red)

- **Click on the "Legacy Routers" tile — it is highlighted with a red border**



Legacy Routers page — "This is a DEMO site only"

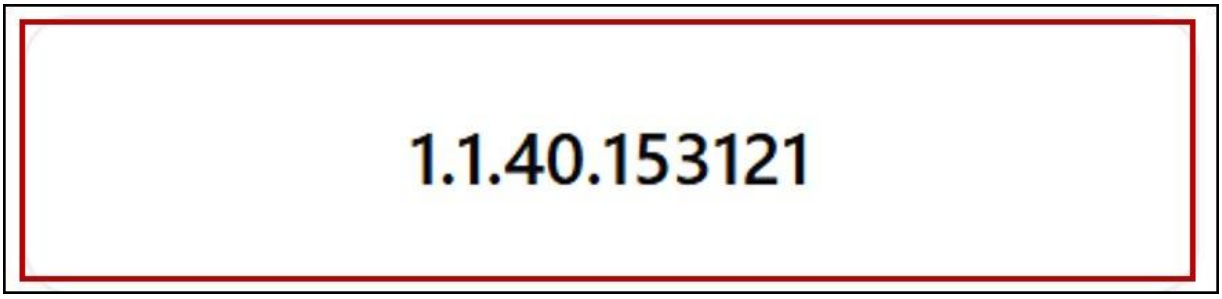
➤ **You are now on the Legacy Routers page — scroll down to find the EA6200**

NOTE: *The breadcrumb at the top shows "Home / Legacy Routers" — confirming you are in the right section.*



Select EA6200 (highlighted in red)

➤ **Click on "EA6200" — it is highlighted with a red border in the image above**

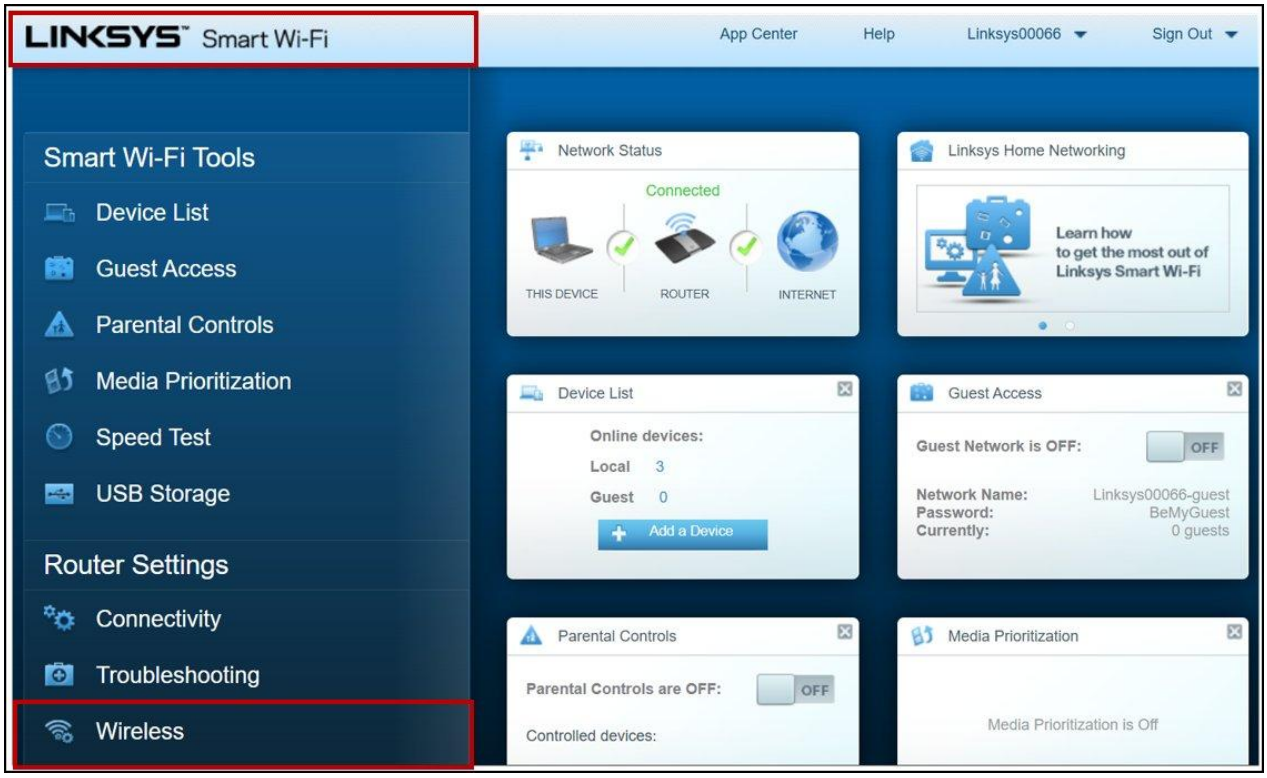


Firmware version shown: 1.1.40.153121

➤ **Note the firmware version 1.1.40.153121 — this confirms you selected the correct router model**

Step 2: Configure Wireless Settings (SSID & WPA2 Encryption)

After the router dashboard loads, click on "Wireless" in the left sidebar under Router Settings. This is where we configure the network name (SSID), password, and encryption type for both bands.

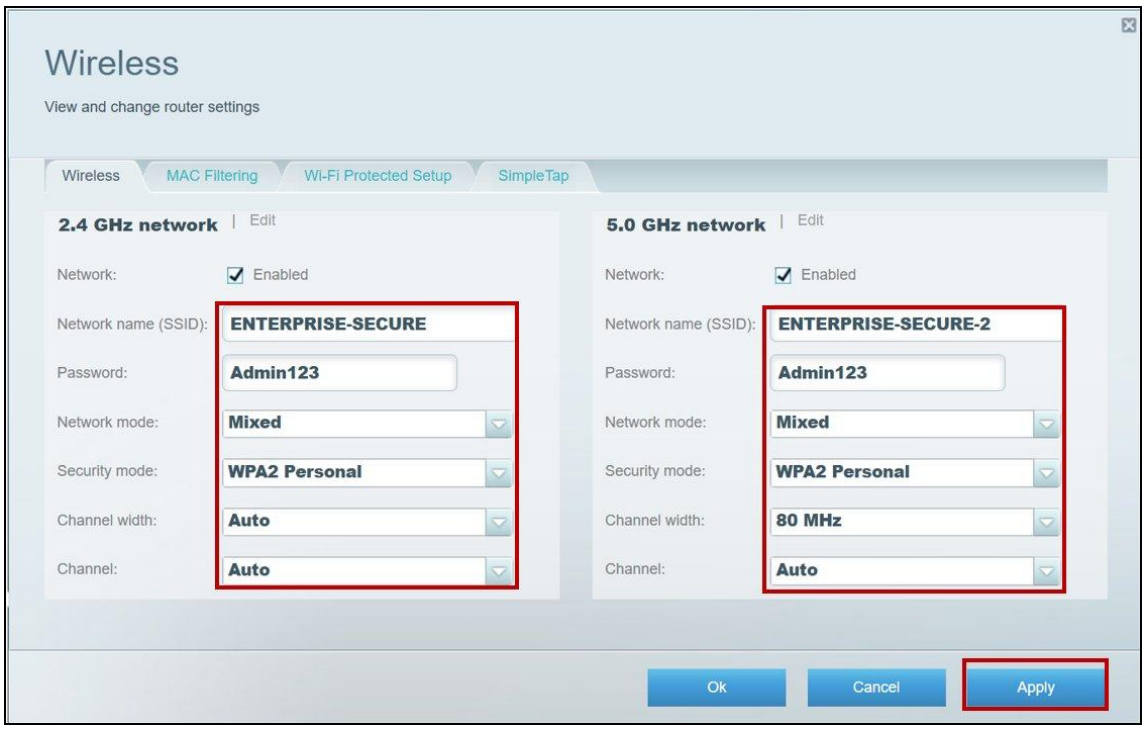


Smart Wi-Fi dashboard — Click "Wireless" in the left sidebar (highlighted in red)

- **From the Smart Wi-Fi dashboard, click "Wireless" in the left sidebar**
- **You will see the Wireless settings page with 2.4 GHz and 5.0 GHz network sections**

Configure SSID, Password & Security Mode

On the Wireless settings page, click "Edit" next to the 2.4 GHz network, and then also for the 5.0 GHz network. Set the values as shown below:



Wireless settings filled in – ENTERPRISE-SECURE, Admin123, WPA2 Personal

- **2.4 GHz Network Name (SSID): type "ENTERPRISE-SECURE"**
- **2.4 GHz Password: type "Admin123"**
- **Security mode: select "WPA2 Personal" from the dropdown**
- **5.0 GHz Network Name (SSID): type "ENTERPRISE-SECURE-2"**
- **5.0 GHz Password: type "Admin123"**
- **5.0 GHz Security mode: select "WPA2 Personal"**
- **Click "Apply" button (bottom right) to save the changes**

NOTE: *WPA2 Personal is the strongest encryption mode for home/small business routers. It uses AES encryption which is far more secure than the older WEP and WPA standards.*

Confirm the Disconnect Warning

After clicking Apply, a Disconnect Warning dialog will appear. This is normal — it warns that connected devices may be disconnected when wireless settings change. Click "Yes" to confirm.



Disconnect Warning — click "Yes" to confirm (highlighted in red)

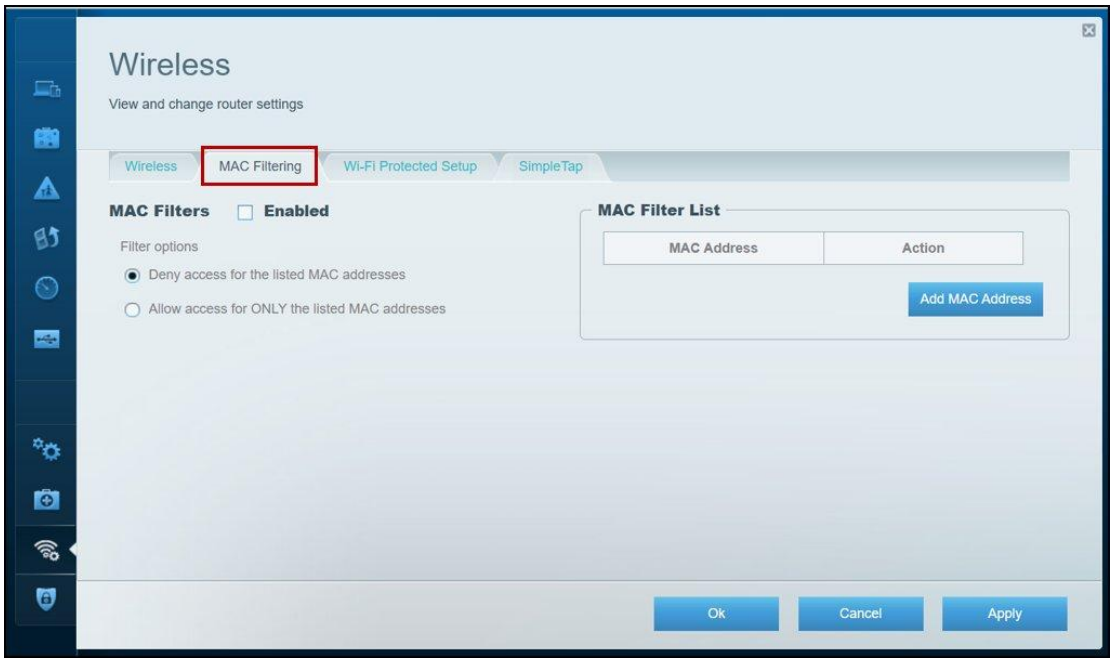
- **Read the warning — it shows the new SSID and security settings for both bands**
- **Click "Yes" button to apply and confirm the changes**

NOTE: *The dialog confirms: 2.4 GHz → ENTERPRISE-SECURE (WPA2 Personal), 5.0 GHz → ENTERPRISE-SECURE-2 (WPA2 Personal). This is correct.*

Step 3: Enable MAC Address Filtering

MAC Address filtering allows us to control which devices can connect to the wireless network. We will enable it in whitelist mode — meaning ONLY devices whose MAC address is on our list will be allowed to connect.

From the Wireless settings page, click on the "MAC Filtering" tab.

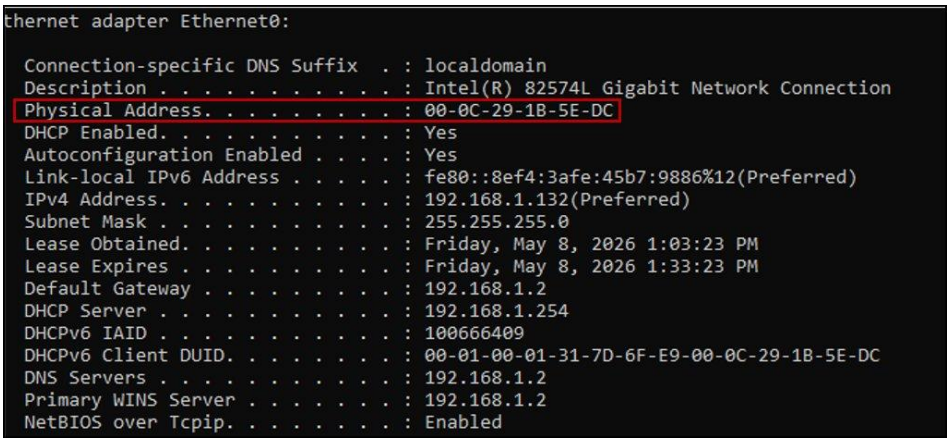


MAC Filtering tab — currently disabled, no MAC addresses in the list

- **Click on the "MAC Filtering" tab at the top of the Wireless settings page**
- **You can see the "MAC Filters" checkbox is currently unchecked (disabled)**

Get Your VM MAC Address

Before enabling MAC filtering, we need the MAC address of the virtual machine (VM). Open Command Prompt on the Windows VM and run "ipconfig /all" to find the Physical Address.

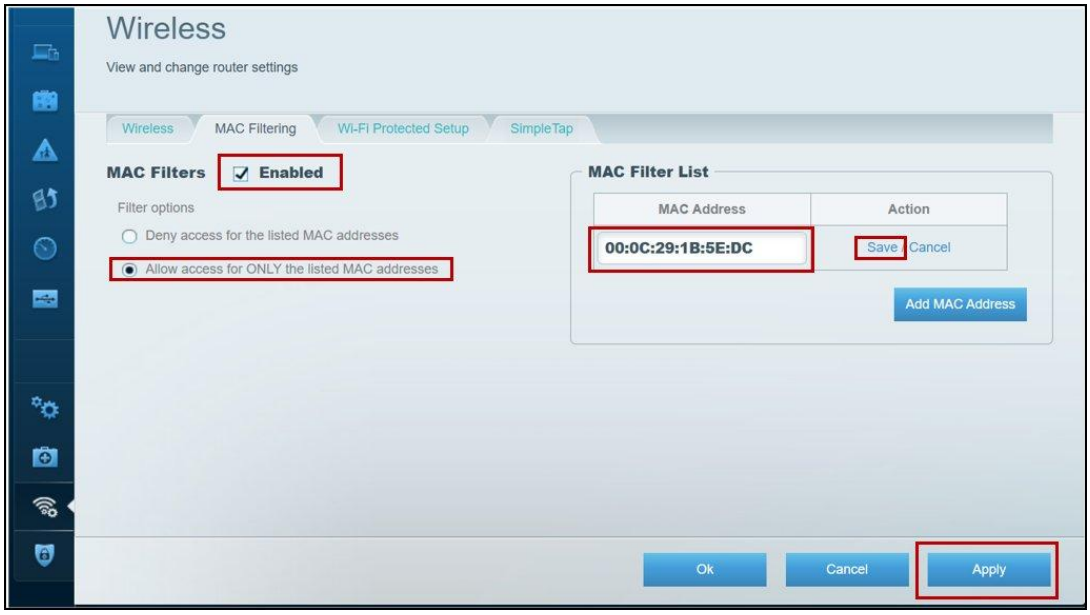


ipconfig /all output — Physical Address: 00-0C-29-1B-5E-DC

- **Open Command Prompt on your Windows VM**
- **Type: ipconfig /all and press Enter**
- **Find "Physical Address" under Ethernet Adapter — note it down**
- **VM MAC Address found: 00-0C-29-1B-5E-DC (will be entered as 00:0C:29:1B:5E:DC)**

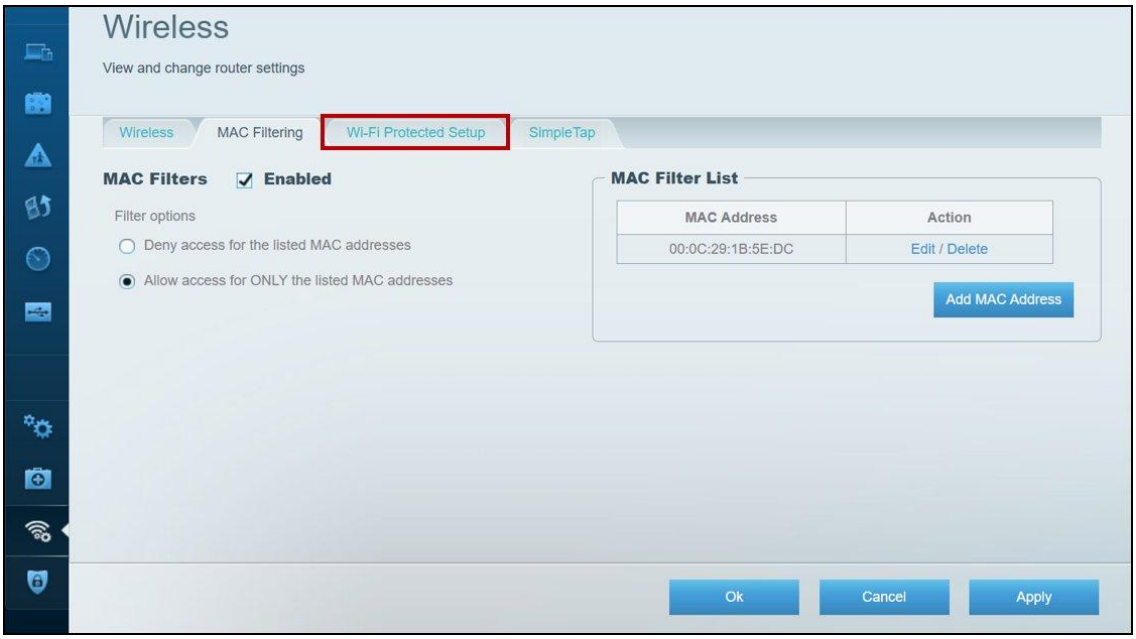
Add MAC Address and Enable Whitelist

Now go back to the MAC Filtering tab, enable it, select whitelist mode, and add the VM MAC address.



MAC Filtering – Enabled, whitelist mode selected, VM MAC address entered

- **Check the "Enabled" checkbox to activate MAC Filtering**
- **Select "Allow access for ONLY the listed MAC addresses" (whitelist mode)**
- **Click "Add MAC Address" button**
- **Enter: 00:0C:29:1B:5E:DC in the MAC Address field**
- **Click "Save" to save the MAC entry**
- **Click "Apply" button to apply all MAC filtering settings**



MAC address saved in list – Edit/Delete option visible, WPS tab highlighted

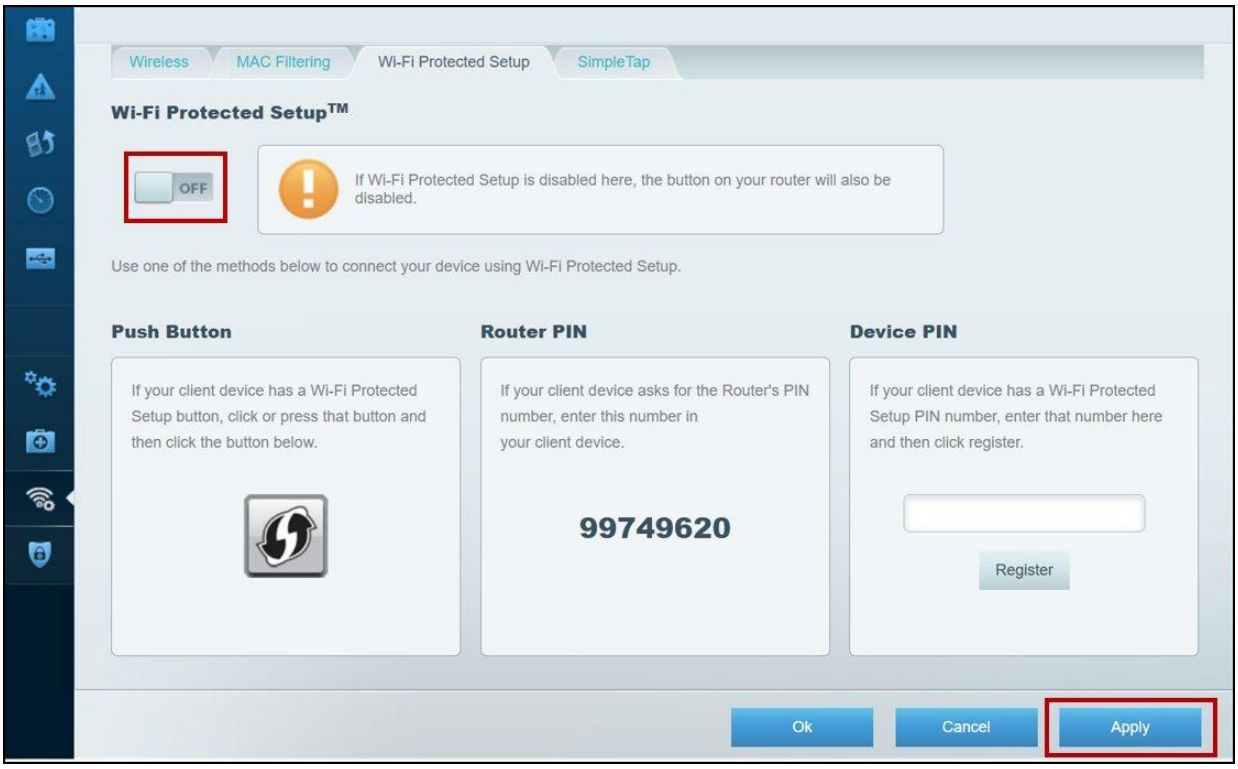
- **The MAC address 00:0C:29:1B:5E:DC now appears in the MAC Filter List**
- **Action column shows "Edit / Delete" – confirming it was saved correctly**

NOTE: *Whitelist mode is more secure than blacklist mode. Only devices in the list can connect – all others are automatically blocked.*

Step 4: Disable WPS (Wi-Fi Protected Setup)

WPS (Wi-Fi Protected Setup) is a convenient feature that lets devices connect easily with a push button or PIN. However, it has known security vulnerabilities and should always be disabled on enterprise networks.

Click on the "Wi-Fi Protected Setup" tab (next to MAC Filtering tab).



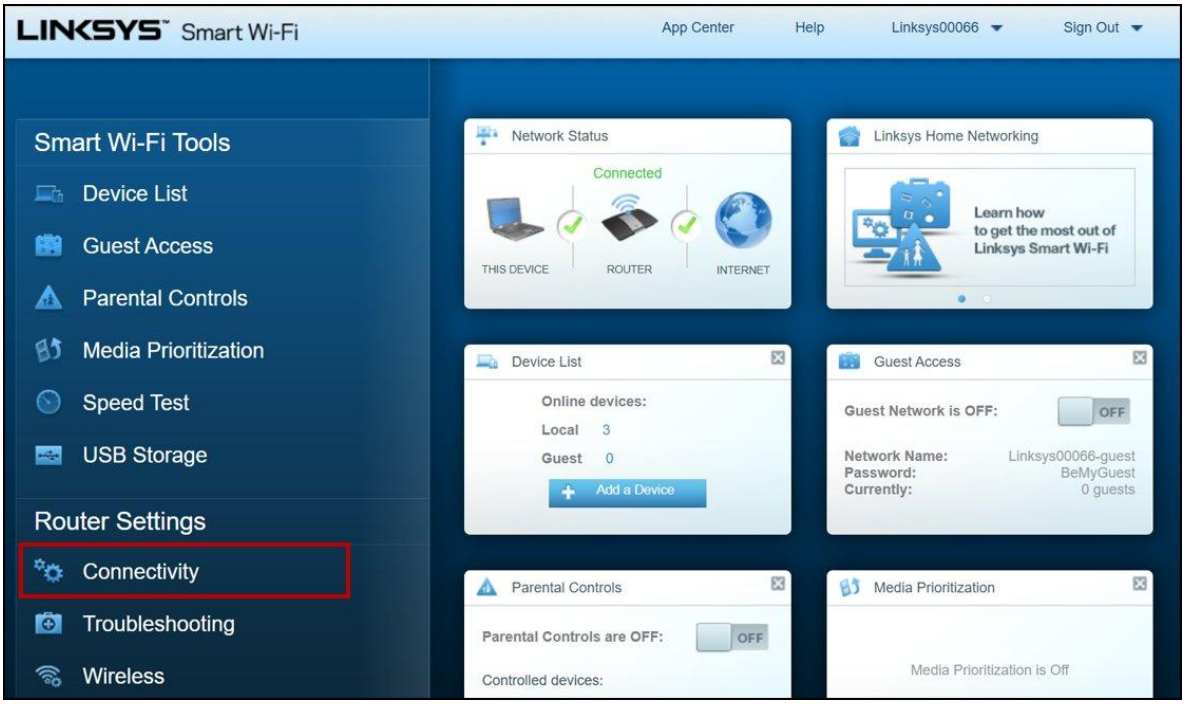
Wi-Fi Protected Setup — toggle set to OFF, Apply highlighted

- **Click on the "Wi-Fi Protected Setup" tab**
- **You will see the WPS toggle — click it to turn it OFF**
- **The toggle should show "OFF" (as shown in the image above)**
- **Click "Apply" button to save the change**

NOTE: *WPS PIN attacks (Pixie Dust, brute-force) can crack WPS in hours. Disabling WPS removes this attack vector entirely. The physical button on the router will also be disabled.*

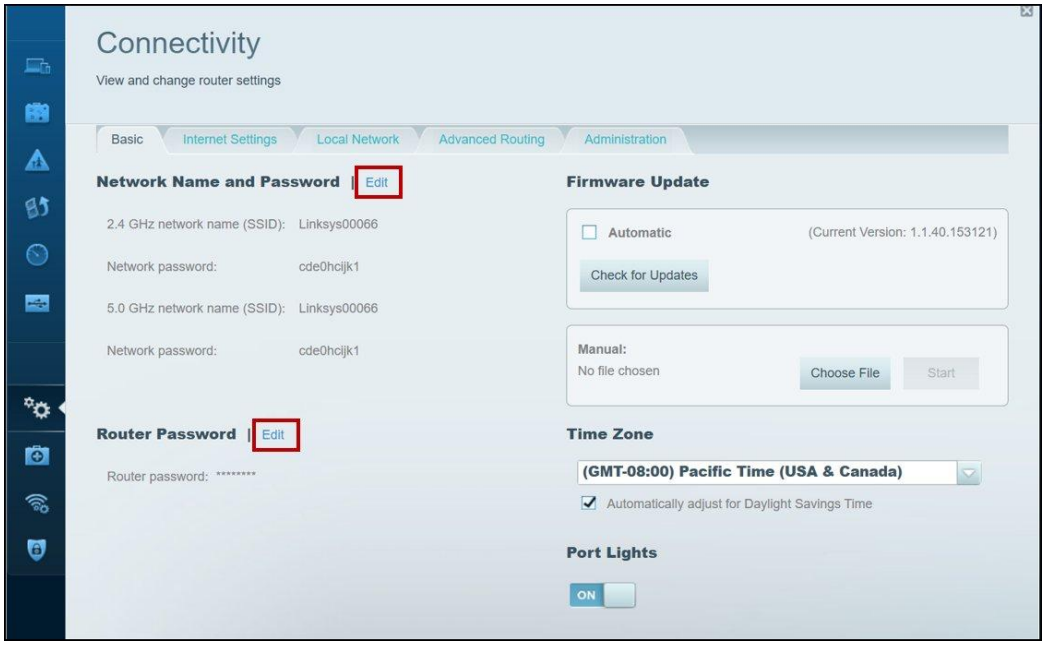
Step 5: Change Router Admin Password & Network Settings

Click "Connectivity" in the left sidebar under Router Settings. This section allows us to change the Router Admin password (the password used to log into the router interface itself).



Smart Wi-Fi dashboard — click "Connectivity" in the left sidebar (highlighted)

➤ **From the main dashboard, click "Connectivity" in the left sidebar**

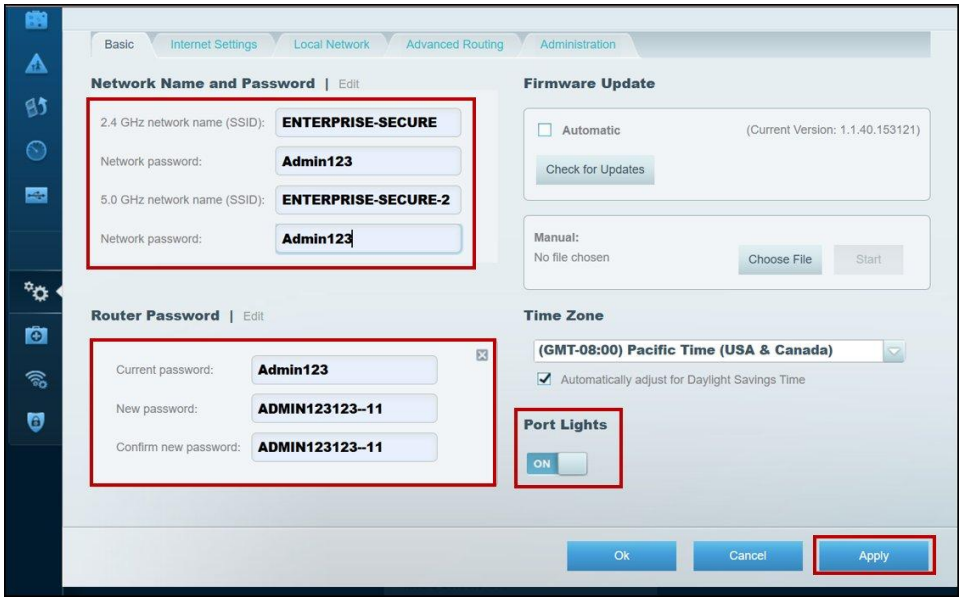


Connectivity → Basic tab — "Edit" buttons highlighted for Network Name and Router Password

- You will see the Connectivity → Basic tab
- Click the "Edit" button next to "Network Name and Password" to update the SSID
- Click the "Edit" button next to "Router Password" to change the admin password

Set New Network Names and Admin Password

In the Edit mode, update the network names and set a new strong router admin password as shown below:



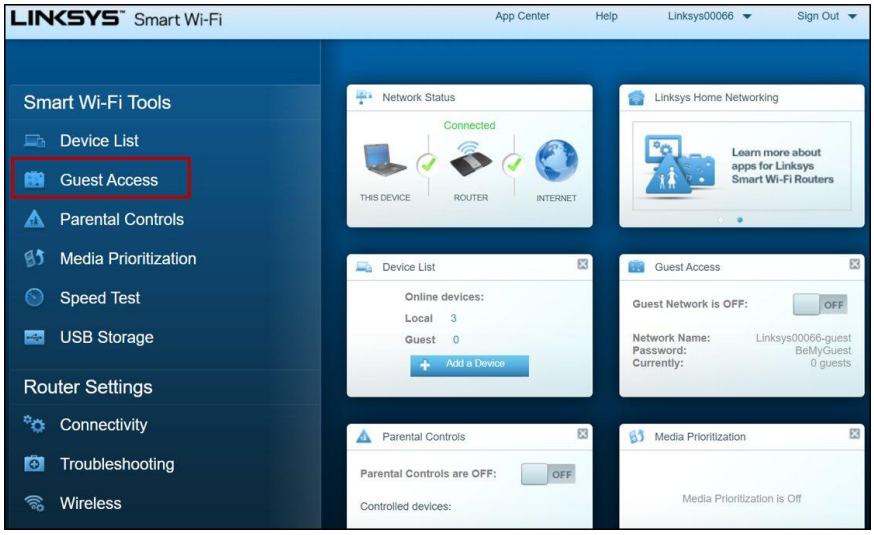
Connectivity Basic — ENTERPRISE-SECURE SSIDs and Admin password change form filled

- 2.4 GHz Network Name: ENTERPRISE-SECURE (already set in previous step)
- 2.4 GHz Network Password: Admin123
- 5.0 GHz Network Name: ENTERPRISE-SECURE-2 (already set in previous step)
- 5.0 GHz Network Password: Admin123
- Current password (Router): Admin123
- New password (Router): ADMIN123123--11
- Confirm new password: ADMIN123123--11
- Port Lights: leave ON (no change needed)
- Click "Apply" button to save all changes

NOTE: *The Router Admin Password (ADMIN123123--11) is separate from the Wi-Fi password. It controls who can access the router settings interface. Always change this from the default.*

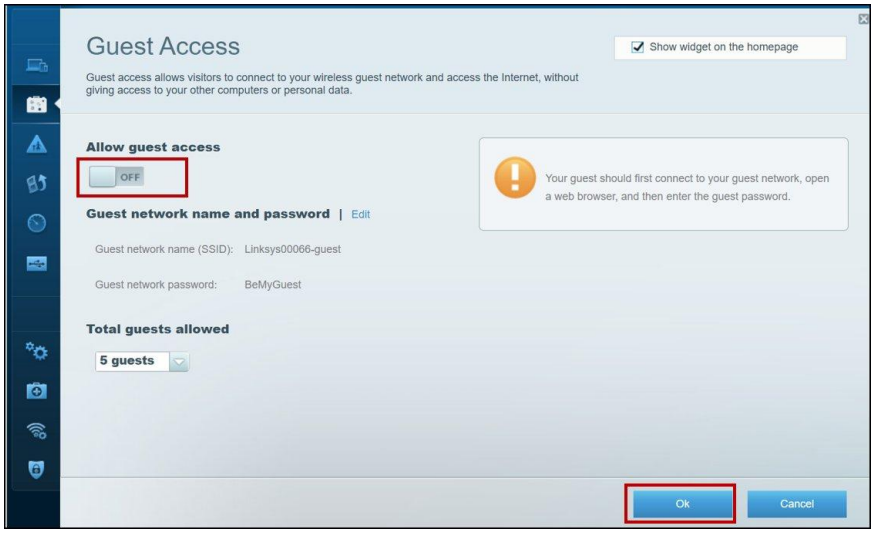
Step 6: Disable Guest Network Access

The Guest Network allows visitors to connect to the internet without accessing the main network. However, for security hardening, we disable it to prevent any unauthorized wireless access.



Smart Wi-Fi dashboard — click "Guest Access" in the left sidebar (highlighted)

➤ **Click on "Guest Access" in the left sidebar under Smart Wi-Fi Tools**



Guest Access page — toggle is OFF, click "Ok" to confirm (highlighted)

➤ **The "Allow guest access" toggle should be set to OFF**

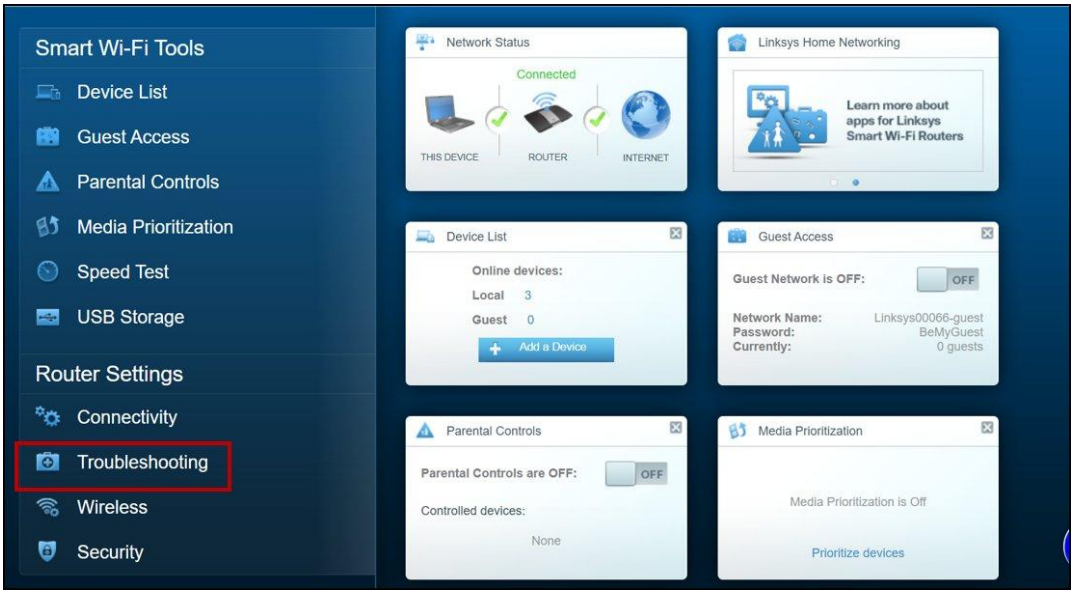
➤ **If it is ON, click the toggle to turn it OFF**

➤ **Click "Ok" button to save and close**

NOTE: *With Guest Access disabled, only devices with the correct WPA2 password AND an approved MAC address can connect to the network.*

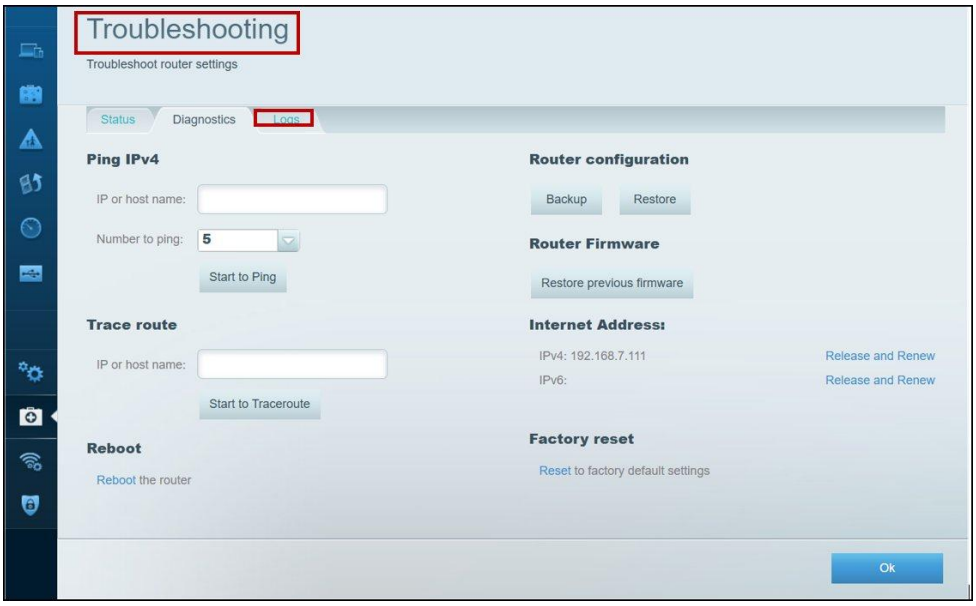
Step 7: Enable Router Logs

Router logs help monitor network activity and detect security events. We enable logging in the Troubleshooting section.



Smart Wi-Fi dashboard — click "Troubleshooting" in the left sidebar (highlighted)

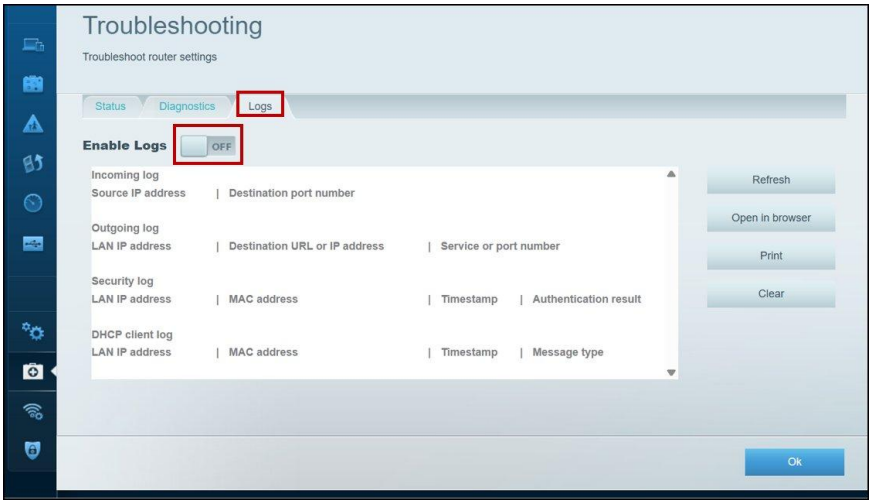
➤ **Click on "Troubleshooting" in the left sidebar under Router Settings**



Troubleshooting — Diagnostics tab open, click on "Logs" tab (highlighted in red)

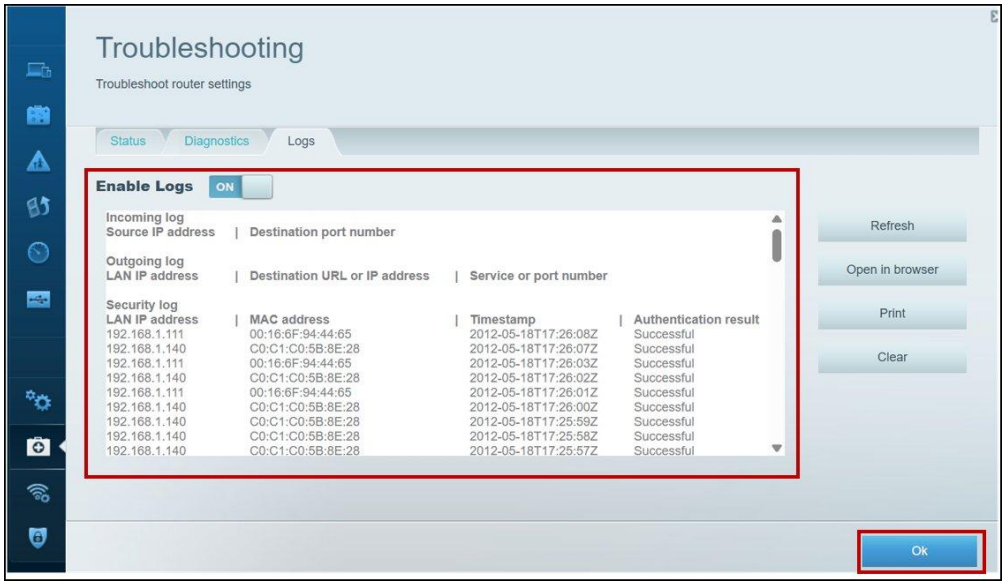
➤ **The Troubleshooting page opens on the Diagnostics tab by default**

➤ **Click on the "Logs" tab at the top**



Troubleshooting → Logs tab — "Enable Logs" toggle is currently OFF

- **You will see the Logs tab with "Enable Logs" toggle currently set to OFF**
- **Click the toggle to turn Enable Logs ON**



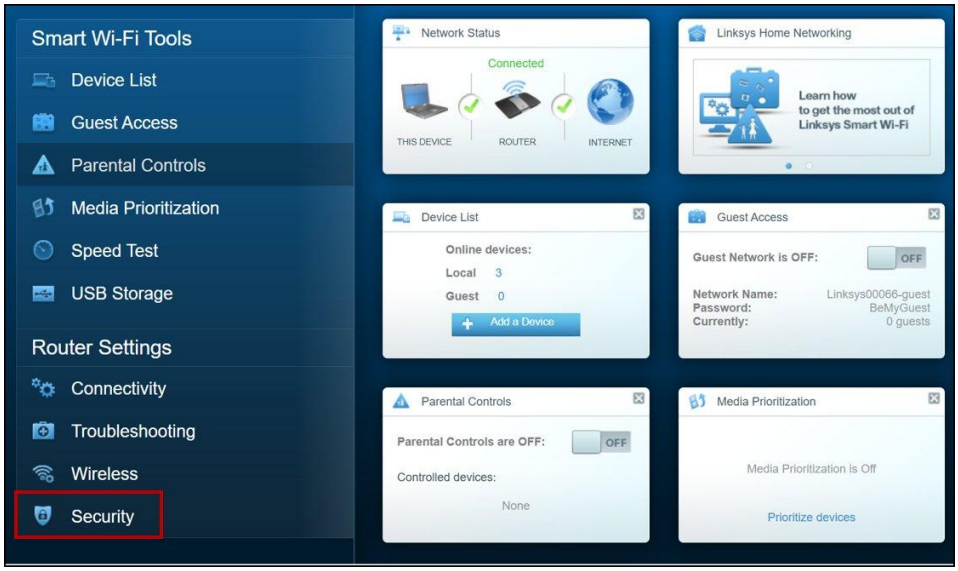
Logs enabled (ON) — Security log shows authentication events with IP, MAC, Timestamp

- **After enabling logs, the toggle shows "ON" and log data appears**
- **You can see the Security log showing: LAN IP address, MAC address, Timestamp, Authentication result**
- **All entries show "Successful" — confirming normal authentication activity**
- **Click "Ok" to close and save**

NOTE: *The Security Log records every authentication attempt to the router. In a real network, these logs would be reviewed regularly to detect unauthorized access attempts.*

Step 8: Verify Firewall & Security Settings

Click on "Security" in the left sidebar to review and verify the firewall configuration. This section contains three tabs: Firewall, DMZ, and Apps and Gaming.

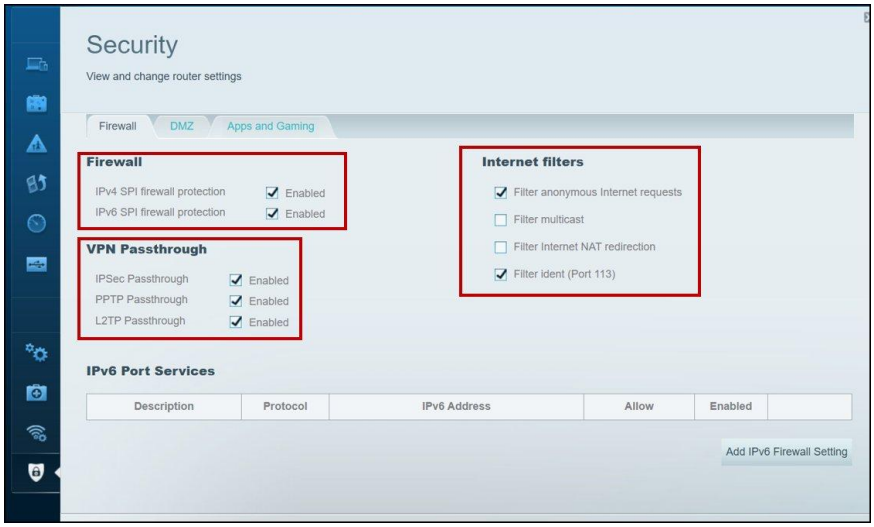


Smart Wi-Fi dashboard — click "Security" in the left sidebar (highlighted)

➤ **Click on "Security" in the left sidebar under Router Settings**

Firewall Tab — Verify All Settings

The Firewall tab shows IPv4 and IPv6 firewall status, VPN passthrough settings, and Internet filters. Verify that the important security options are enabled:



Security → Firewall tab — IPv4/IPv6 SPI Firewall ENABLED, Internet filters configured

➤ **Firewall section: confirm IPv4 SPI firewall protection = Enabled (checkbox checked)**

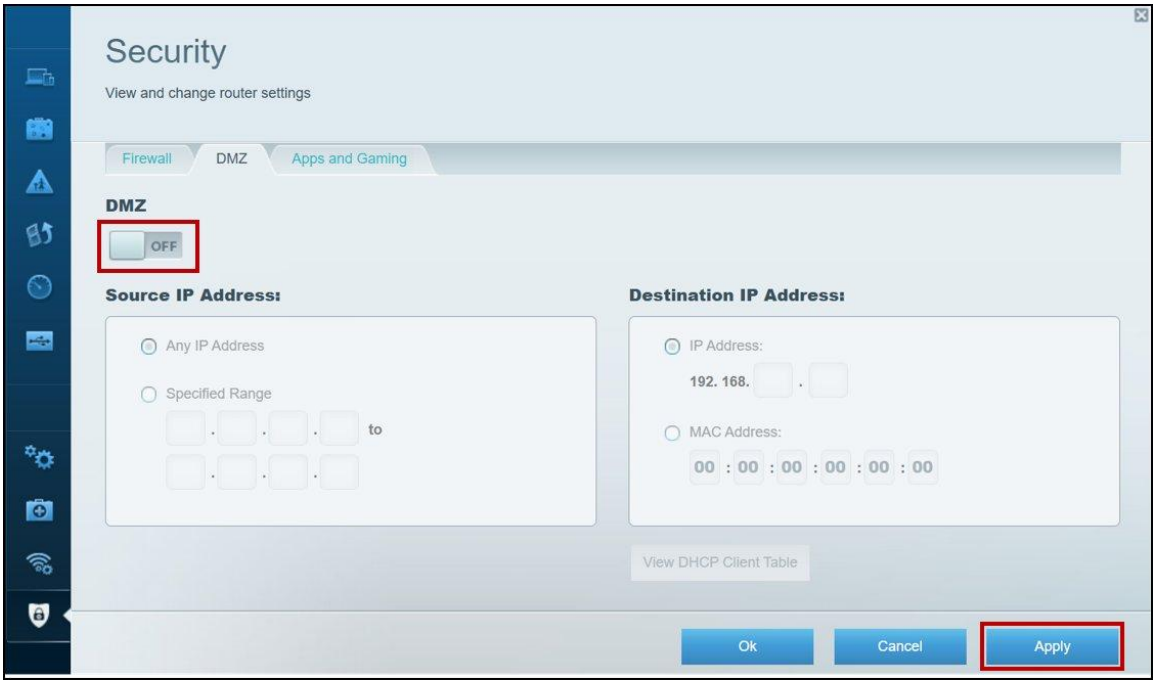
➤ **Firewall section: confirm IPv6 SPI firewall protection = Enabled (checkbox checked)**

- **Internet Filters: "Filter anonymous Internet requests" = CHECKED (enabled)**
- **Internet Filters: "Filter multicast" = leave unchecked**
- **Internet Filters: "Filter Internet NAT redirection" = leave unchecked**
- **Internet Filters: "Filter ident (Port 113)" = CHECKED (enabled)**
- **VPN Passthrough: IPSec, PPTP, and L2TP all = Enabled (for legitimate VPN users)**

> **NOTE:** *The SPI (Stateful Packet Inspection) firewall examines incoming traffic and blocks unauthorized packets. "Filter anonymous Internet requests" prevents the router from responding to ping/port scans from the internet.*

DMZ Tab — Ensure DMZ is Disabled

DMZ (Demilitarized Zone) exposes a device directly to the internet with no firewall protection. This must remain disabled unless specifically needed.



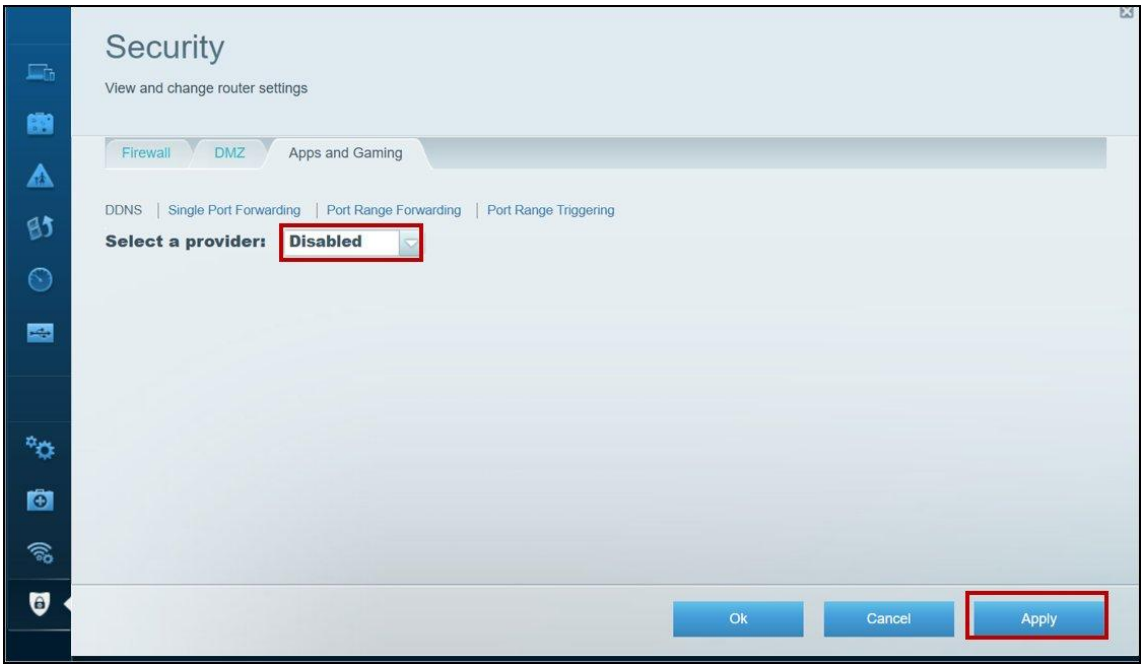
Security → DMZ tab – DMZ toggle is OFF (highlighted in red)

- **Click on the "DMZ" tab**
- **Confirm the DMZ toggle is set to OFF — do NOT enable this**
- **No Source or Destination IP addresses should be configured**
- **Click "Apply" to confirm**

> **NOTE:** *DMZ places a device outside the firewall, exposing it to all internet traffic. Keeping DMZ disabled protects all devices on the network.*

Apps and Gaming Tab — Verify DDNS is Disabled

The Apps and Gaming tab includes DDNS (Dynamic DNS), port forwarding, and port triggering settings. For security hardening, ensure DDNS is disabled.



Security → Apps and Gaming → DDNS tab – Provider set to "Disabled" (highlighted)

- **Click on the "Apps and Gaming" tab**
- **Ensure the DDNS tab is selected**
- **Confirm "Select a provider" dropdown shows "Disabled"**
- **Click "Apply" to confirm**

> **NOTE:** *Disabling DDNS prevents the router from broadcasting its external IP address to dynamic DNS providers, reducing the network's external visibility.*

Lab Completion Summary

All wireless security hardening tasks have been successfully completed on the Linksys EA6200 router simulator. The table below summarizes every security configuration applied during this lab:

Security Feature	Configuration Applied	Status
Wireless Encryption	WPA2 Personal (AES)	☑ Secured
2.4 GHz SSID	ENTERPRISE-SECURE	☑ Renamed
5.0 GHz SSID	ENTERPRISE-SECURE-2	☑ Renamed
Network Password	Admin123 (demo)	☑ Changed
Router Admin Password	ADMIN123123--11	☑ Hardened
MAC Filtering	Whitelist — VM MAC added	☑ Enabled
WPS (Wi-Fi Protected Setup)	Disabled	☑ Secured
Guest Network	Disabled	☑ Off
IPv4 SPI Firewall	Enabled	☑ On
IPv6 SPI Firewall	Enabled	☑ On
Filter Anonymous Internet Requests	Enabled	☑ On
Filter Ident (Port 113)	Enabled	☑ On
DMZ	Disabled	☑ Off
DDNS	Disabled	☑ Off
Router Logs	Enabled	☑ On

Key Security Concepts Demonstrated

- **— strongest wireless encryption for personal/SMB use**
- **— only pre-approved devices can connect**
- **— removes PIN/button-based connection attack vectors**
- **— eliminates secondary access point**
- **— prevents unauthorized router management access**
- **— stateful packet inspection on both IP versions**
- **— hides router from internet ping scans**
- **— no devices exposed directly to the internet**
- **— no external hostname mapping**
- **— security audit trail for all authentication events**
